1.5 Applications of Quadratic Equations

Quadratic equations often appear in problems involving geometry, motion, and optimization. In this section, we use the methods learned so far to solve practical problems involving volume, area, and the Pythagorean Theorem.

Definition

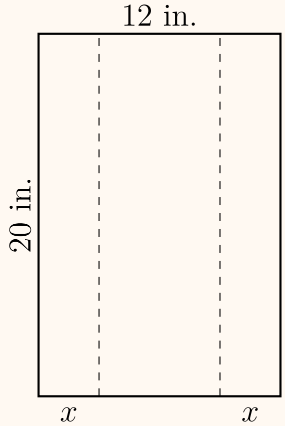
A **quadratic model** is any equation that can be expressed in the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

and describes a real-world relationship such as area, volume, or motion.

Real-World Example

Solving an Application Involving Volume
A rectangular piece of aluminum measuring 20 inches long and 12 inches wide is folded along each long edge to form a trough. The sides are folded up x inches. If the volume of the trough is to be 360 in^3, how far from the edge should the fold be made?



Solution:

$$V = 20(12 - 2x)x = 360$$

$$40x^2 - 240x + 360 = 0 \Rightarrow (x-3)^2 = 0$$

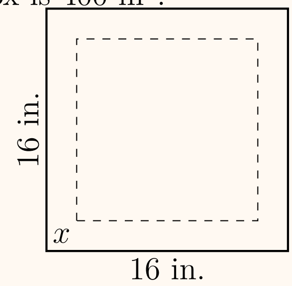
$$x = 3$$

Answer: The folds should be made 3 inches from each edge.

Real-World Example

Constructing an Open Box from Sheet Metal

A square sheet of metal measuring 16 inches on each side is used to make an open box by cutting out squares of side x from each corner and folding up the sides. Find x if the volume of the box is 400 in^3.



Solution:

$$V = x(16 - 2x)^2 = 400$$
$$x(256 - 64x + 4x^2) = 400$$
$$4x^3 - 64x^2 + 256x - 400 = 0$$

Dividing by 4:

$$x^3 - 16x^2 + 64x - 100 = 0$$

Testing $x = 2$ gives 0, so $x = 2$ in. **Answer:** Cut 2 in. squares from each corner.

Real-World Example

Finding Box Dimensions from Volume

The width of a rectangular box is half its height. The length is 5 cm longer than its height. If the volume is 150 cm^3, find the dimensions.

Solution: Let height $= x$. Then:

$$\text{width} = \tfrac{1}{2}x, \quad \text{length} = x + 5$$

$$V = (x)(\tfrac{1}{2}x)(x + 5) = 150$$
$$\tfrac{1}{2}x^3 + \tfrac{5}{2}x^2 - 150 = 0$$

Multiply by 2:

$$x^3 + 5x^2 - 300 = 0$$

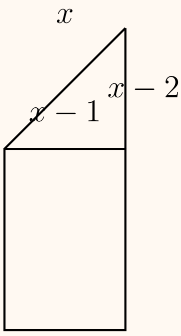
Testing $x = 10$ gives 0, so height $x = 10$ cm.

$$\text{Width} = 5 \text{ cm}, \quad \text{Length} = 15 \text{ cm}.$$

Real-World Example

Solving an Application Involving the Pythagorean Theorem

A window consists of a rectangle topped by a right triangle. One leg of the triangle is 2 ft shorter than the hypotenuse, and the other leg is 1 ft shorter. Find the lengths of the sides.



Solution:

$$(x-1)^2 + (x-2)^2 = x^2$$

$$x^2 - 6x + 5 = 0 \Rightarrow (x-5)(x-1) = 0$$

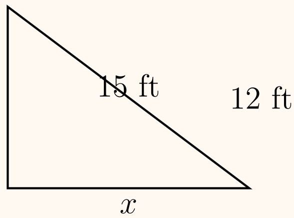
$$x = 5$$

Answer: Legs are 4 ft and 3 ft, hypotenuse 5 ft.

Real-World Example

Ladder Against a Wall

A 15-ft ladder is placed so that its base is x ft away from a wall and the top touches the wall at a height of 12 ft. How far is the base from the wall?



Solution:

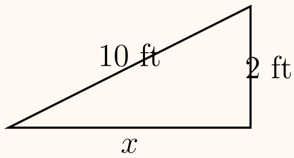
$$x^2 + 12^2 = 15^2 \Rightarrow x^2 = 81 \Rightarrow x = 9$$

Answer: The base is 9 ft from the wall.

Real-World Example

Building a Ramp

A wheelchair ramp must rise 2 ft over a horizontal distance of x ft. If the ramp length must be 10 ft, find x.



Solution:

$$x^2 + 2^2 = 10^2 \Rightarrow x^2 = 96 \Rightarrow x \approx 9.8$$

Answer: The horizontal distance is approximately 9.8 ft.

Real-World Example

Analyzing an Object Moving Vertically

A model rocket is launched upward from a platform 1 m high with an initial velocity of 24 m/s. The height after t seconds is

$$s(t) = -4.9t^2 + 24t + 1$$

a. When is it at 20 m?

Solution:

$$-4.9t^2 + 24t + 1 = 20 \Rightarrow 4.9t^2 - 24t + 19 = 0$$

$$t = \frac{24 \pm \sqrt{576 - 372.4}}{9.8} \approx 0.9, 4.3$$

Answer: $t \approx 0.9$ s (up), $t \approx 4.3$ s (down).

Real-World Example

Throwing a Ball Upward

A ball is thrown upward from ground level with an initial velocity of 32 ft/s. The height (in ft) after t seconds is

$$h = -16t^2 + 32t$$

Find when the ball hits the ground.

Solution:

$$-16t^2 + 32t = 0 \Rightarrow t(-16t + 32) = 0$$

$$t = 0 \text{ or } t = 2$$

Answer: The ball hits the ground after 2 seconds.

Real-World Example

Dropping an Object from a Height
An object is dropped from a 60 m building. The height after t seconds is

$$h = -4.9t^2 + 60$$

Find when it hits the ground.
Solution:

$$-4.9t^2 + 60 = 0 \Rightarrow t^2 = \frac{60}{4.9} \Rightarrow t \approx 3.5$$

Answer: The object hits the ground after about 3.5 seconds.

Extra Practice

1. The length of a rectangle is 3 yd more than twice its width. The area is 629 yd^2.
2. The width of a rectangle is 2 m less than one-quarter its length. The area is 252 m^2.
3. The height of a triangle is 2 ft less than the base. The area is 40 ft^2.
4. The height of a triangle is 4 yd longer than the base. The area is 70 yd^2.
5. The width of a rectangular box is 8 in. The height is one-fifth its length, and the volume is 640 in^3.
6. The height of a rectangular box is 4 ft. The length is 1 ft longer than twice its width. The volume is 312 ft^3.
7. The longer leg of a right triangle is 2 ft longer than the shorter leg, and the hypotenuse is 2 ft shorter than twice the shorter leg.
8. The longer leg of a right triangle is 7 cm longer than the shorter leg, and the hypotenuse is 17 cm.